

The Generalized Lubrication Equation as a subgrid contact-line model: theory, numerics, and validation

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Abstract

A moving contact line carries a viscous stress that diverges at the solid, so the interface shape below the smallest resolved scale of a direct numerical simulation (DNS) cannot be captured on the DNS mesh. This document sets out a subgrid treatment of that region built on the Generalized Lubrication Equation (GLE) of Snoeijer, which stays quantitative for steep interface slopes where classical lubrication theory fails. We record the derivation and sign conventions of the arc-length GLE system, the two solvers that share its right-hand side (adaptive Cash–Karp shooting and a fixed-mesh collocation branch tracer), the calibration used to reproduce the dip-coating bifurcation diagram of Snoeijer & Andreotti, the numerical verification of every component, and the seam that couples the GLE to a Basilisk two-phase DNS at the grid scale. The whole solver stack is header-only C99 with no external dependencies.

1 Introduction

The hydrodynamic model of a moving contact line predicts a shear stress that grows like $1/r$ towards the contact line, so the viscous dissipation carries a non-integrable logarithm and no solution exists within continuum hydrodynamics alone [1]. A microscopic mechanism must cut the singularity off; the simplest and the one used here is a Navier slip length λ , below which the no-slip condition is relaxed. The physical consequence is that the interface bends over a range of scales set by λ at the bottom and by the macroscopic geometry (here the capillary length $\ell_\gamma = \sqrt{\gamma/\rho g}$) at the top — six or more decades of arc length for a millimetric bath and a nanometric slip length.

Resolving all of those decades on a single DNS mesh is infeasible. The strategy adopted here divides the labour by length scale. A subgrid model carries the viscous bending of the interface from the slip length up to the DNS grid size Δ ; a Basilisk two-phase DNS carries it from Δ up to the macroscopic scale. The two are coupled at Δ : each DNS time step the subgrid model hands the DNS an *apparent* contact angle at the grid scale, and takes the DNS interface curvature at that scale as its own outer boundary condition. The subgrid wedge never has to be resolved on the DNS mesh.

The subgrid model is the GLE of Snoeijer [4], a long-wavelength theory that — unlike ordinary lubrication theory — remains quantitative when the interface is steep, and reduces to lubrication theory in the small-slope limit. This document covers, in turn: the derivation and conventions of the GLE in the arc-length form used in the code (Section 2); the two numerical solvers and the calibration to the dip-coating problem (Section 3); the verification of each component against independent references (Section 4); and the GLE $\leftrightarrow$ DNS coupling seam (Section 5).

2 Theory

2.1 From the wedge expansion to the generalized lubrication equation

Classical lubrication theory expands the Stokes equations in the interface slope and is therefore restricted to $|\nabla h| \ll 1$. Snoeijer’s observation is that at low capillary number the physical requirement is not that the interface be flat, but that its *curvature* vary slowly: surface tension is strong enough to keep the shape close to that of a straight wedge, whatever the opening angle [4]. One therefore expands the Stokes flow about the Huh–Scriven similarity solution for a wedge of finite, slowly varying opening angle $\theta(x)$ [1], treating the variation of θ as the perturbation. The small parameter is $\varepsilon = \text{Ca}^{1/2}$, with $\text{Ca} = \eta U^*/\gamma$, and the expansion is self-consistent as long as

$$h \partial_x \theta \ll 1, \quad (1)$$

i.e. the interface angle turns over a length long compared with the local film thickness. There is no restriction on the size of θ itself.

The result is the generalized lubrication equation (Snoeijer [4], Eq. 14),

$$\partial_x \kappa = 3 \text{Ca} \frac{U/U^*}{h^2} F(\theta) + \frac{1}{\gamma} \partial_x \Phi|_{z=h}, \quad \kappa = \frac{\partial_{xx} h}{(1 + \partial_x h^2)^{3/2}}, \quad (2)$$

with the full curvature κ retained (not its small-slope approximation), the body-force potential Φ , and the mobility factor (Snoeijer [4], Eq. 15)

$$F(\theta) = \frac{2}{3} \frac{\tan \theta \sin^2 \theta}{\theta - \cos \theta \sin \theta}. \quad (3)$$

As $\theta \rightarrow 0$, $F(\theta) \rightarrow 1$ and Eq. (2) collapses onto the ordinary lubrication equation, so the GLE is a slope-dependent correction of the viscous term that is exact at leading order in ε for any slope.

2.2 The arc-length system

Parameterising the interface by its curvature keeps Eq. (2) single-valued through overhangs and folds, which the Cartesian form cannot represent. Let s be the arc length along the liquid–gas interface, measured from the contact line, and take the state vector $y = (h, \theta, \omega, \zeta)$, with h the film thickness normal to the plate, θ the local interface inclination, $\omega = \text{d}\theta/\text{d}s$ the curvature, and ζ the distance travelled along the plate. The GLE becomes the first-order system

$$\frac{\text{d}h}{\text{d}s} = \sin \theta, \quad \frac{\text{d}\theta}{\text{d}s} = \omega, \quad \frac{\text{d}\omega}{\text{d}s} = \frac{3 \text{Ca} M(\theta, \mu_r)}{h(h + c_\lambda \lambda)} + G(\theta), \quad \frac{\text{d}\zeta}{\text{d}s} = \cos \theta. \quad (4)$$

The denominator $h(h + c_\lambda \lambda)$ is the Navier-slip regularisation of the bare h^2 : it removes the contact-line singularity of the viscous term while recovering h^2 for $h \gg \lambda$. The cutoff coefficient $c_\lambda = c(\theta_e, \mu_r)$ follows from Cox–Voinov matching with slip [8]. The value $c_\lambda = 3$ is the classical one-fluid, small-angle convention, not a universal constant. The implementation accepts an explicit coefficient for arbitrary (θ_e, μ_r) and includes the closed-form right-angle result; a general-angle evaluator would additionally require solving Hocking’s integral equation and is deliberately not guessed.

2.3 The two-fluid Huh–Scriven mobility

For a passive outer fluid the mobility is $F(\theta)$ of Eq. (3). The code carries the full two-fluid Huh–Scriven wedge factor, which accounts for viscous dissipation in the displaced phase at a gas/liquid viscosity ratio μ_r [7]:

$$M(\theta, \mu_r) = \frac{2 \sin^3 \theta [\mu_r^2 f_1(\theta) + 2\mu_r f_3(\theta) + f_1(\pi - \theta)]}{3 [\mu_r f_1(\theta) f_2(\pi - \theta) + f_1(\pi - \theta) f_2(\theta)]}, \quad (5)$$

with

$$f_1(\theta) = \theta^2 - \sin^2 \theta, \quad f_2(\theta) = \theta - \sin \theta \cos \theta, \quad f_3(\theta) = \theta(\pi - \theta) + \sin^2 \theta. \quad (6)$$

The *plus* sign in the denominator of Eq. (5) is load-bearing: a minus sign there flips the sign of the entire viscous term at small μ_r , and every earlier C port in this repository's history carried that error. In the one-fluid limit ($\mu_r = 0$) the general form reduces to

$$M(\theta, 0) = \frac{2 \sin^3 \theta}{3 f_2(\theta)} = \cos \theta F(\theta), \quad (7)$$

and $M \rightarrow 1$ as $\theta \rightarrow 0$, recovering the classical result. The mobility is not named $\mathbf{f}$ in the code because in Basilisk $\mathbf{f}$ is the volume-of-fluid volume fraction.

2.4 Sign conventions and the flat-film consistency check

We take $\text{Ca} = \eta U / \gamma > 0$ for a *receding* contact line — the dip-coating configuration in which the plate is withdrawn from the bath — and $\text{Ca} < 0$ for an advancing line. The consistency check is the flat-film limit of the vertical plate: setting $\theta \rightarrow 0$ and $\omega' \rightarrow 0$ in Eq. (4) balances the viscous term against gravity and gives the entrained-film thickness

$$h_{\text{film}} = \sqrt{3 \text{Ca}} \ell_\gamma, \quad (8)$$

the well-known result recovered by Snoeijer [4] (Fig. 3 of that paper) for a film dragged down a vertical plate against a receding contact line.

Gravity enters through

$$G(\theta) = -g^* \cos \theta \quad (\text{vertical plate}), \quad G(\theta) = +g^* \sin \theta \quad (\text{horizontal plate}), \quad (9)$$

with $g^* = (\ell_\gamma / \ell_{\text{unit}})^2$ the gravity prefactor: $g^* = 1$ when lengths are measured in capillary lengths, and $g^* = 0$ disables gravity (the length unit is then free). Retaining g^* as a general scaling rather than fixing it to unity makes the gravity-rescaling invariance of Section 4 an explicit test rather than an identity. In the vertical geometry s runs from the contact line *down* towards the bath, so the elevation is $z(s) = \Delta - \zeta(s)$ and the static far field obeys the hydrostatic balance $\omega = g^* z$.

2.5 The oscillatory film tail and small negative angles

On the upper (film) branch of the dip-coating problem the interface joins the meniscus through an oscillatory tail, the classical dimple oscillation of a Landau–Levich film, in which θ dips slightly below zero. The mobility must therefore be continued to small negative angles, and how it is continued depends on μ_r . In the one-fluid limit $M(\theta, 0) = 2 \sin^3 \theta / [3 f_2(\theta)]$ is analytically *even* in θ , so the exact even extension $M(\theta) = M(|\theta|)$ applies, with $M(0) = 1$. For $\mu_r > 0$ the mobility is *not* even: near $\theta = 0$ it carries a genuine linear term,

$$M(\theta, \mu_r) = 1 + \frac{3 \mu_r \theta}{2\pi} + \mathcal{O}(\theta^2), \quad (10)$$

so the raw two-fluid formula is evaluated at the *signed* θ (the apparent 0/0 at $\theta = 0$ is removable and set explicitly to $M = 1$). Because f_1 and f_2 lose all significant digits to cancellation for small arguments ($f_1 \sim \theta^4/3$, $f_2 \sim 2\theta^3/3$), both switch to their Taylor series below $\theta = 0.02$, which is also more accurate than the direct trigonometric form there.

2.6 The dip-coating boundary-value problem

The steady meniscus at a withdrawn plate is a two-point boundary-value problem for the system (4). At the inner boundary, the contact line,

$$h(s_0) = h_0, \quad \theta(s_0) = \theta_e, \quad \omega(s_0) = \omega_0 \text{ (unknown)}, \quad (11)$$

with θ_e the microscopic (equilibrium) contact angle. The integration starts at a small but finite $s_0 \sim \lambda$ rather than at $s = 0$ because the curvature diverges logarithmically at the contact line even with slip; the contribution of the interval $(0, s_0)$ is a bounded logarithm absorbed into the calibration.

The outer boundary condition is that the trajectory has landed on the static meniscus manifold by the matching thickness $h = H$. Static solutions ($\text{Ca} = 0$, vertical plate) conserve the first integral $\frac{1}{2}\omega^2 + g^* \sin \theta$; the branch connected to a flat bath ($\omega \rightarrow 0$, $\theta \rightarrow \pi/2$) therefore satisfies

$$\omega|_{h=H} = \sqrt{2g^*(1 - \sin \theta|_{h=H})}. \quad (12)$$

Matching to Eq. (12) at a moderate H (the code uses $H = 5$) is legitimate because the viscous term in Eq. (4) has by then decayed like $3\text{Ca} M/h$. The meniscus rise follows from the hydrostatic balance $\omega = g^*z$ on the manifold,

$$\Delta = \zeta|_{h=H} + \omega|_{h=H}/g^*, \quad (13)$$

and the apparent contact angle from the classical meniscus-rise relation $\Delta = \sqrt{2(1 - \sin \theta_{\text{app}})/g^*}$ (Snøeijer & Andreotti [9], Eq. 4), i.e.

$$\theta_{\text{app}} = \arcsin\left(1 - \frac{1}{2}g^*\Delta^2\right). \quad (14)$$

The branch $\Delta(\text{Ca})$ turns at a saddle-node fold at Ca^* : there the apparent angle vanishes, $\theta_{\text{app}} \rightarrow 0$, and the rise attains its critical value. Setting $\theta_{\text{app}} = 0$ in Eq. (14) gives $\Delta = \sqrt{2/g^*}$, i.e.

$$z_c = \sqrt{2} \ell_\gamma \quad (15)$$

in capillary-length units. Above Ca^* no steady meniscus exists and a film is entrained: the Landau–Levich transition, at which the upper branch approaches the entrained-film thickness of Eq. (8). The existence of this maximum speed, and the avoided-critical-behaviour structure around it, were established for this system by Eggers [10] and by Snøeijer *et al.* [5]; the experimental branch is that of Delon *et al.* [6].

3 Numerical implementation

Two complementary solvers share the right-hand side (4). Both are header-only C99 and need only `libm` — no GSL, no SUNDIALS — so the same headers compile under a host compiler and under Basilisk’s `gcc`.

3.1 Adaptive Cash–Karp integration

The trajectory spans the slip scale $\lambda \sim 10^{-6}\ell_\gamma$ to the capillary scale in one integration, and near the contact line $d\omega/ds \sim 1/s$, which a fixed step cannot follow. The integrator is an embedded Cash–Karp Runge–Kutta 5(4) pair with proportional step control. The step error is measured in the mixed norm

$$E = \left[\frac{1}{4} \sum_{i=1}^4 \left(\frac{\text{err}_i}{\text{atol} + \text{rtol} \max(|y_i|, |y_i^5|)} \right)^2 \right]^{1/2}, \quad (16)$$

a step is accepted when $E \leq 1$, and the next step is scaled by $\min(5, \max(0.2, 0.9 E^{-1/5}))$; the same comparison rejects a NaN error norm, which fails every ordinary inequality. Integration terminates on a *thickness event* $h = H$, localised to a relative tolerance of 10^{-13} on h by bisection on the final step size. This is how the outer matching point of the boundary-value problem is hit exactly.

3.2 Single shooting on the contact-line curvature

The static-manifold condition (12) reduces the boundary-value problem to a single scalar equation in the unknown ω_0 ,

$$\mathcal{R}(\omega_0) \equiv \omega|_{h=H} - \sqrt{2g^*(1 - \sin\theta|_{h=H})} = 0, \quad (17)$$

solved by a damped Newton iteration with a central finite-difference derivative, falling back to geometric bracket expansion and bisection on stagnation. Trajectories that leave the physical domain before reaching H return signed penalties chosen so that a collapsing interface ($\theta \rightarrow 0$, $h \rightarrow 0$) reads as ω_0 too small and a fold-over ($\theta \rightarrow \pi$) as ω_0 too large; bisection then still brackets.

The conditioning of Eq. (17) sets a floor on the achievable residual. The flat-bath fixed point of Eq. (4) has a growing mode, so a perturbation at the contact line is amplified downstream like e^s . The matching therefore cannot be pushed to large H without losing all precision, and even at $H = 5$ the residual floor with the default integrator tolerances is $\sim 10^{-8}$; the Newton tolerance is set at 5×10^{-8} accordingly. (The alternative far-field condition $\omega(s_{\max}) = 0$, used only for the cross-validation of Section 4, has $\mathcal{O}(1)$ sensitivity in ω and supports a tolerance of 10^{-11} .) Shooting is the fast single-solve path and the engine of the DNS coupling; on the lower branch it is reliable up to roughly 0.6 Ca^* , beyond which the residual window in ω_0 narrows exponentially towards the fold.

3.3 Collocation branch tracing through the fold

Shooting cannot trace the full bifurcation diagram. In the (ω_0, Ca) plane the branch makes a hairpin at the fold that is narrower than the shooting map can resolve: the forward flow *compresses* the separation between neighbouring branches like e^{-s} while amplifying noise like e^{+s} . Deep on the upper branch the trajectory carries a film section of length $\sim \Delta$ whose exponential dichotomy makes shooting ill-conditioned in *both* directions. The chart (ω_0, Ca) is therefore degenerate at the fold.

The branch tracer instead discretises the whole problem on a *fixed* mesh and parameterises solutions by the meniscus rise Δ , which is monotone along the entire branch. Nodes $s_i = s_0 (s_{\text{end}}/s_0)^{\tau_i}$ use a hybrid grading, logarithmic below $s = 0.3$ (to resolve the slip-to-mesoscale decades near the contact line) and uniform above (to resolve the film and meniscus foot, whose widths are $\mathcal{O}(\sqrt{\text{Ca}})$); only the uniform part dilates with the unknown domain length s_{end} . The cell residuals are the second-order implicit midpoint rule,

$$y_{i+1} - y_i - (s_{i+1} - s_i) \mathbf{f}\left(\frac{1}{2}(y_i + y_{i+1})\right) = 0, \quad (18)$$

closed by the contact-line conditions, the static-manifold condition (12) at node N , the definition $h_N = H$ (which fixes s_{end}), and the target rise $\zeta_N + \omega_N/g^* = \Delta^*$. The unknowns are the $4(N+1)$ node states together with Ca and s_{end} ; the equations are the $4N$ collocation residuals plus six conditions. The system is square.

The Jacobian is block-bidiagonal of bandwidth seven with two dense border columns ($\partial/\partial \text{Ca}$ and $\partial/\partial s_{\text{end}}$, formed by finite differences of the full residual) and two border rows (the h_N and Δ conditions). It is solved by the bordering algorithm: banded LU with partial pivoting on the square band part, three back-solves, and a 2×2 reduced system for the two border unknowns.

Because Δ is monotone along the branch, marching Δ^* with a secant predictor and an adaptive step (halved on a failed corrector, grown by $1.4\times$ on fast convergence) walks straight through the fold with no turning-point handling at all. This is exactly what the mesh re-adaptation of SciPy’s `solve_bvp` inside a continuation loop could not do — the failure mode of the historical Python continuation (issues #14 and #16 of the repository) — because a mesh that moves between Newton iterations makes the extended continuation system inconsistent. A fixed mesh keeps it consistent. The construction is a stabilised multiple-shooting march in the sense of local parameterisation continuation [15].

3.4 Parameters and calibration for Fig. 4b

The Fig. 4b reproduction has two fitted microscopic inputs, the equilibrium angle θ_e and an effective slip parameter λ , with the legacy cutoff convention fixed explicitly at $c_\lambda = 3$. They are calibrated against the vector-digitised theory curve of Fig. 4b of Snoeijer & Andreotti [9], not measured independently:

- $\theta_e = 53.46^\circ$, fixed by the static foot $\Delta(\text{Ca} \rightarrow 0) = \sqrt{2(1 - \sin \theta_e)} = 0.627$;
- $\lambda/\ell_\gamma = 7.46 \times 10^{-6}$, fixed by the digitised fold $\text{Ca}^* = 10.54 \times 10^{-3}$. The prediction is only logarithmically sensitive to λ , so the fold calibration absorbs the chosen $c_\lambda = 3$ cutoff convention into this fitted value. It must not be read as an independently measured physical Navier length.

Because the microscopic cutoff is calibrated *to* Ca^* , recovering Ca^* from the traced branch is circular by construction and is not a test. The genuine out-of-sample checks are the fold height $\Delta^* = 1.440$ (digitised 1.446, not fit to) and the full branch shape, including the upper-branch approach to the Landau–Levich asymptote, which carries no further free parameters.

4 Validation

Each component is verified against an independent reference; the cross-solver and cross-language checks establish that both solvers integrate the same physics, and the mesh-convergence and gravity-invariance checks establish that the discretisation is correct and unit-consistent.

Mobility. The two-fluid mobility (5) agrees with the Python reference to machine precision across $\theta \in (0, \pi)$ and $\mu_r \in [0, 1]$. Below $\theta = 0.02$ the C series form is in fact *more* accurate than the direct trigonometric formula, which loses precision to the cancellation in f_1 and f_2 .

Slip cutoff. The right-angle matching helper reproduces the analytic one-fluid value

$$c(\pi/2, 0) = \exp(\ln 2 - \gamma_E) = 1.1229189671.$$

It also gives the finite-viscosity-ratio values $c(\pi/2, 1) = 2.05$ and $c(\pi/2, 10) = 6.40$ reported by Chan *et al.* [8]. Values away from $\theta_e = \pi/2$ remain an explicit input rather than an undocumented approximation.

Interface profile. Against the historical SciPy `solve_bvp` reference solving the identical boundary-value problem, the interface angle $\theta(s)$ agrees to a relative error below 2.3×10^{-4} throughout the well-conditioned window. The far field is a genuinely soft condition: the gain $d\theta(10^6)/d\omega_0 \sim 5 \times 10^7$, so the two solutions may drift apart in θ at very large s even though the two right-hand sides agree to machine precision. This is a property of the boundary-value problem, not of either implementation.

Shooting versus collocation. At equal Ca the two solvers agree to $\sim 2 \times 10^{-6}$, confirming that the fixed-mesh march and the adaptive shot converge to the same solution branch.

Mesh convergence of the fold. Refining the collocation mesh at $N = 1250, 2500, 5000$ cells gives the fold location

N	Ca^*
1250	1.0543847×10^{-2}
2500	1.0543572×10^{-2}
5000	1.0543504×10^{-2}

The successive differences fall by a factor of about four as the mesh spacing halves, confirming the second-order convergence of the implicit-midpoint discretisation.

Linear algebra. The bordered banded-LU solve agrees with a dense reference factorisation to $\sim 10^{-14}$, and the Cash–Karp tableau reproduces its published coefficients exactly.

Gravity-rescaling invariance. Rescaling the gravity prefactor to $g^* = 4$ should halve both the slip length and the meniscus rise (in the new length unit) while leaving Ca invariant, since Ca is dimensionless. The traced branch reproduces this to 4.5×10^{-6} , a direct test that the general g^* scaling is carried consistently through both solvers.

Static limit. The $\text{Ca} \rightarrow 0$ limit reproduces the static meniscus exactly, as it must, since the outer condition (12) is the static first integral.

Figure 1 shows the traced branch overlaid on the digitised reference, and Fig. 2 the C-versus-Python cross-validation.

5 The DNS and subgrid GLE coupling

The GLE resolves the contact-line region from the slip length up to the DNS grid size Δ ; the Basilisk two-phase DNS resolves everything from Δ up to the macroscopic scale. The coupling is a per-time-step event that exchanges one scalar in each direction at the grid scale. Each step (`event (i++)`):

1. the DNS interface position and curvature κ_{DNS} are sampled in the finest interfacial cell nearest the plate, including an interface aligned exactly with cell faces;
2. the subgrid GLE is solved on $s \in [s_0, s_\Delta]$ with $h(s_\Delta) = \Delta$, the inner contact-line conditions, and the outer condition $\omega(s_\Delta) = -\kappa_{\text{DNS}}$, by shooting on ω_0 . The minus sign follows from $h = y$, $\zeta = -x$ and the phase orientation used by Basilisk’s `curvature()` in this case;
3. the resulting apparent angle $\theta_{\text{app}} = \theta(s_\Delta)$ is imposed on the DNS through the height-function `contact_angle()` boundary condition. The height vector is associated explicitly with the VOF tracer, so Basilisk actually applies the dynamic angle.

This is the multiscale strategy of the review [9] applied as a grid-scale boundary condition, in the spirit of the mesh-dependent dynamic contact angle of Afkhami, Zaleski & Bussmann [12]: the imposed angle is a property of the resolved scale, not a fixed material constant.

The seam is implemented and exercised by a bounded Basilisk runtime regression. It starts from the exactly grid-aligned interface that previously evaded the strict volume-fraction filter, confirms refinement to the intended local grid size, and checks that the GLE updates the

associated contact angle. During a normal run the capillary number is formed from the plate speed relative to the measured contact-line speed, the actual local cell size is passed to the GLE, and the dynamic angle is stored in a dumped scalar for restart. A grid-convergence study of the coupled curvature and contact-line-position samples remains the problem-specific production qualification. A closed-form flux theory such as that of Luo & Gao [14] offers a further out-of-sample target for the coupled model. The linearised GLE has separately been checked against the reference data of Kansal *et al.* [13].

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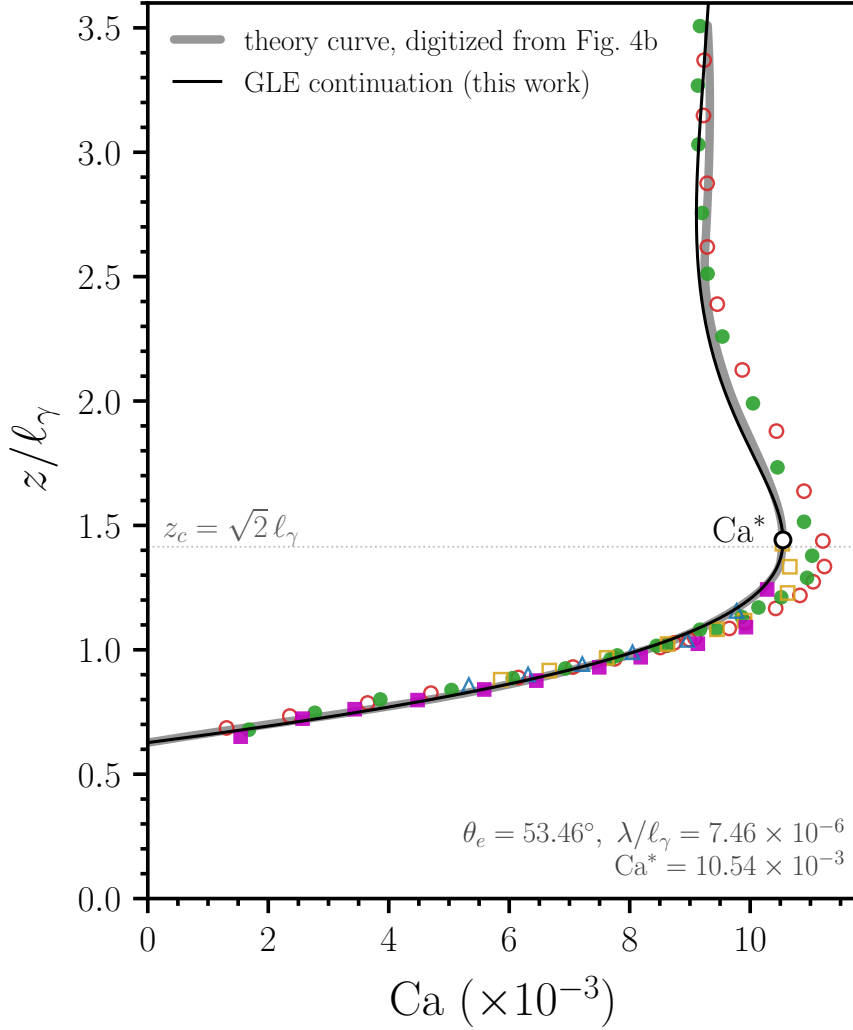


Figure 1: Steady meniscus rise at a receding contact line. Normalised contact-line elevation z/ℓ_γ against capillary number $Ca = \eta U_p/\gamma$ for a partially wetting plate withdrawn from a silicone-oil bath. The black line is the present GLE continuation; the thick grey line is the multiscale-lubrication prediction digitised from Fig. 4b of Snoeijer & Andreotti (2013) [9]. Coloured symbols are the experimental realizations of Delon et al. (2008) [6]: each colour is a distinct run, and the markers denote steady menisci measured at fixed plate speed. In the original review figure these symbols are shadowed by thin continuous coloured traces; those are neither theory nor guides to the eye, but the continuously recorded transient elevation $z_{cl}(t)$ of each run mapped into this plane through the instantaneous relative capillary number $\widehat{Ca}(t) = \eta(U_p - dz_{cl}/dt)/\gamma$. Such transients relax adiabatically through the succession of quasi-steady states and therefore retrace the stationary branch; their visible wiggle is measurement noise amplified by the dz_{cl}/dt term. Only the discrete steady-state symbols are digitised here. The branch folds at Ca^* : the apparent contact angle vanishes, $\theta_{app} \rightarrow 0$, and the rise attains $z_c \rightarrow \sqrt{2} \ell_\gamma$; above Ca^* no steady meniscus exists and a film is entrained.

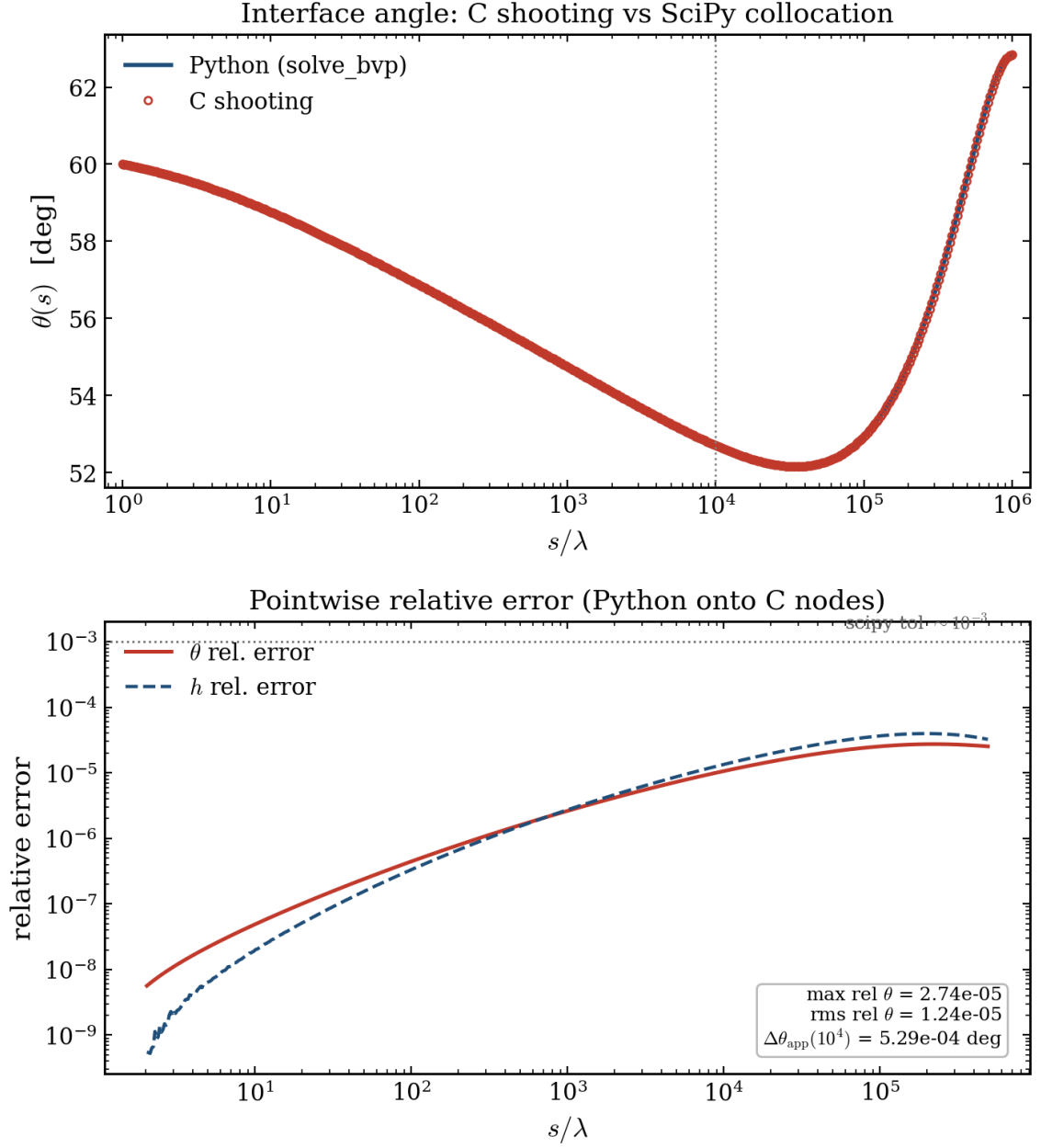


Figure 2: Cross-validation of the C solver against the historical Python `solve_bvp` reference on the identical far-field boundary-value problem. Left: the interface angle $\theta(s)$ on a logarithmic arc-length axis. Right: the relative error between the two solutions on log-log axes. Agreement is excellent where the problem is well conditioned ($s \lesssim 10^4$) and degrades monotonically towards the arc-length cap, a consequence of the soft far-field angle condition rather than of either implementation.